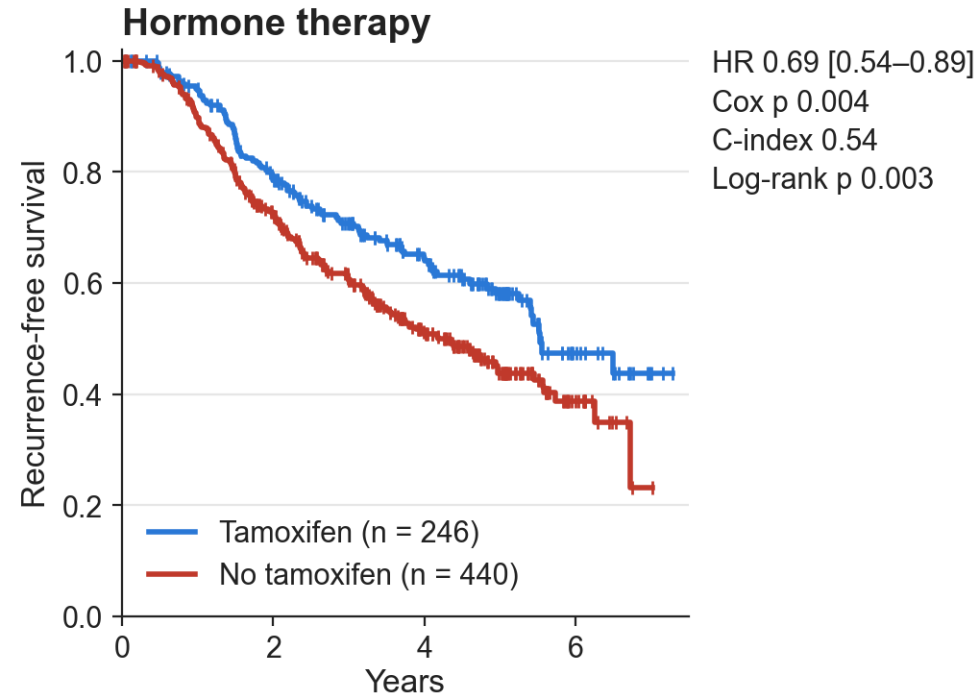


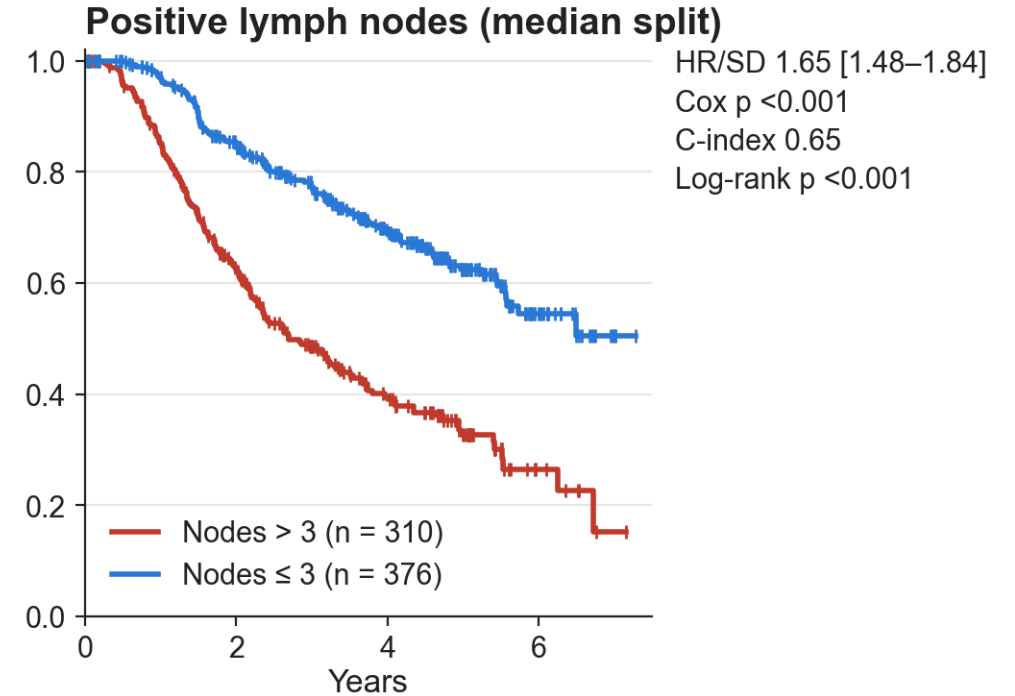
Recurrence-free Survival

Pattern: assertion–evidence



Number at risk

Tamoxifen	246	177	104	18
No tamoxifen	440	281	124	18



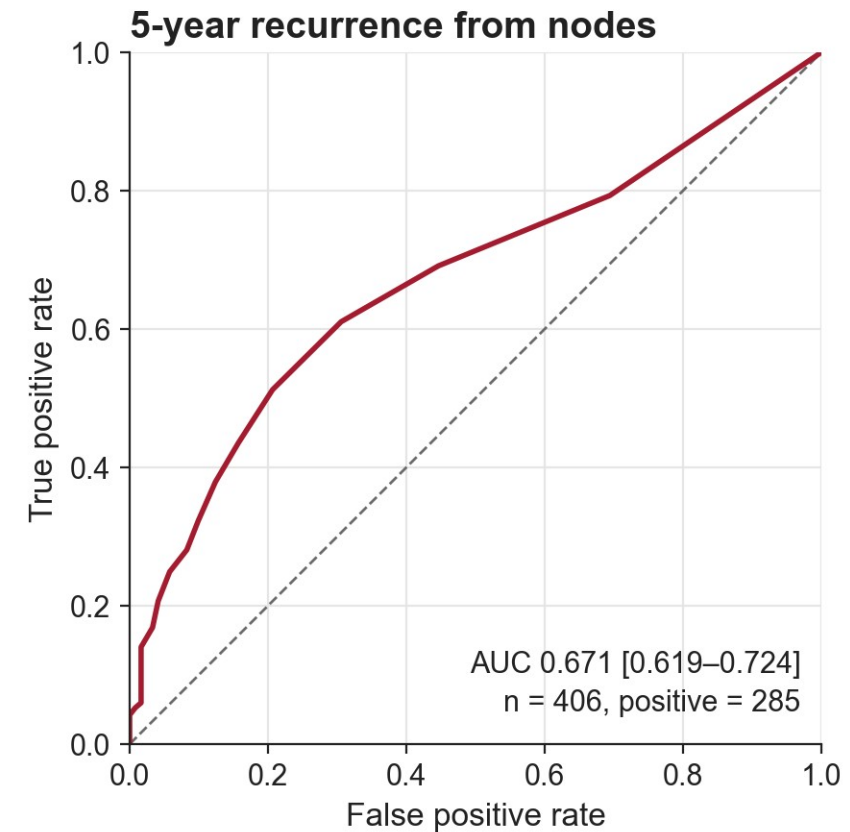
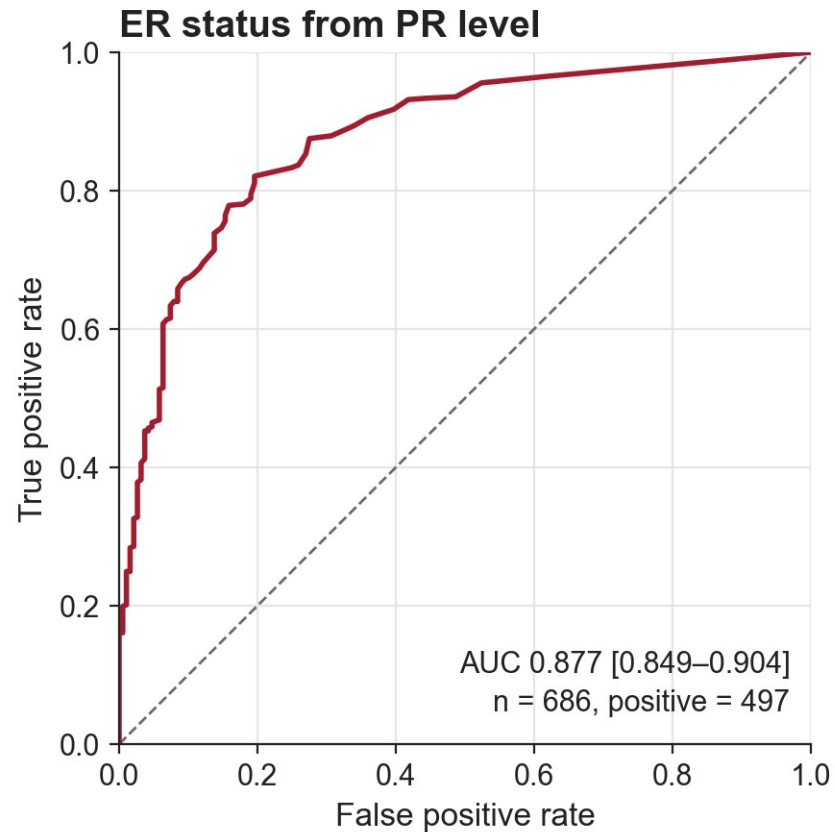
Number at risk

Nodes > 3	310	170	72	8
Nodes ≤ 3	376	288	156	28

Tamoxifen is associated with longer, and more positive nodes with shorter, recurrence-free survival (HR 0.69 [0.54–0.89], p = 0.004; HR 1.65 [1.48–1.84] per SD, p < 0.001; n = 686, 299 events).

Discrimination — ROC

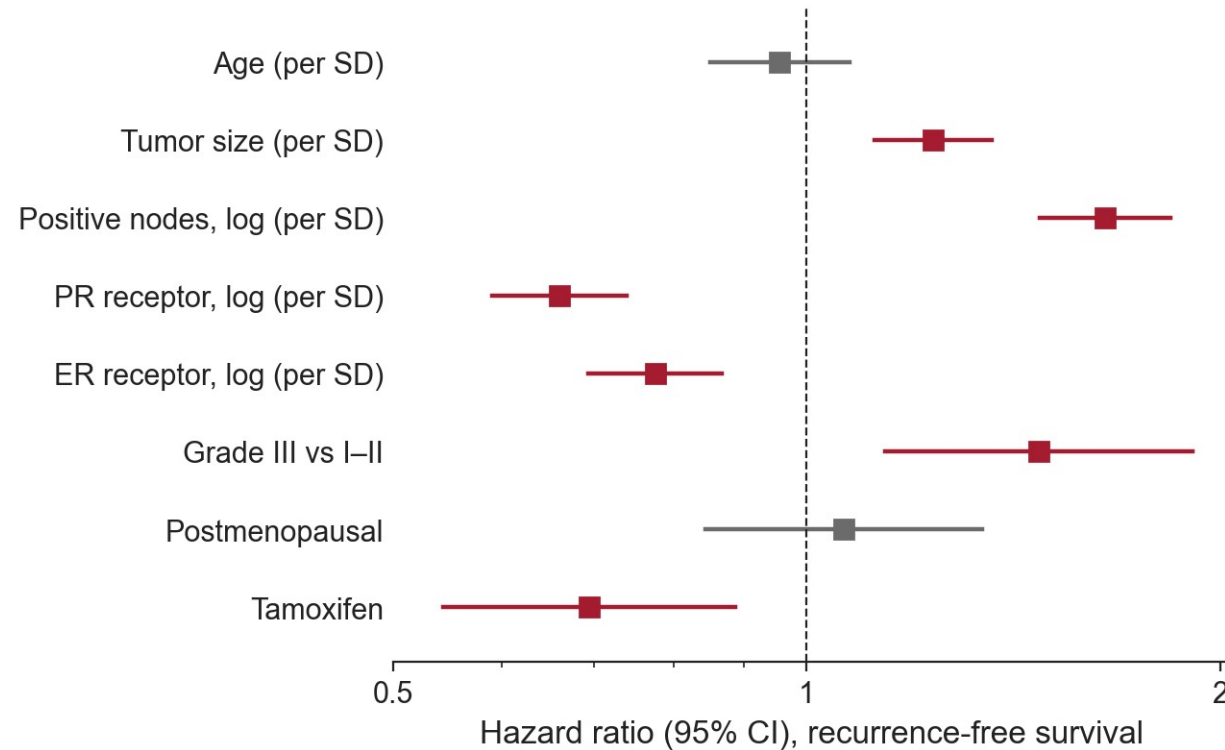
Pattern: two panels, one conclusion



PR level separates ER status well (AUC 0.877 [0.849–0.904]; 497 of 686 ER-positive), whereas nodes predict 5-year recurrence only moderately (AUC 0.671 [0.619–0.724]; 285 of 406 recurred).

Clinical Factors — Univariate Cox

Pattern: forest plot



HR [95% CI]	p	q	n
0.96 [0.85–1.07]	0.446	0.510	686
1.24 [1.12–1.36]	<0.001	<0.001	686
1.65 [1.48–1.84]	<0.001	<0.001	686
0.66 [0.59–0.74]	<0.001	<0.001	686
0.78 [0.69–0.87]	<0.001	<0.001	686
1.48 [1.14–1.91]	0.003	0.005	686
1.06 [0.84–1.34]	0.596	0.596	686
0.69 [0.54–0.89]	0.004	0.005	686

More positive lymph nodes are most strongly associated with shorter recurrence-free survival (HR 1.65 [1.48–1.84] per SD, $q < 0.001$; $n = 686$, 299 events).

Cohort Characteristics

Pattern: native three-line table

Characteristic	All n = 686	Tamoxifen n = 246	No tamoxifen n = 440
Age, years, median (IQR)	53 (46–61)	58 (50–63)	50 (45–59)
Postmenopausal, n (%)	396 (58%)	187 (76%)	209 (48%)
Tumor size, mm, median (IQR)	25 (20–35)	25 (20–35)	25 (20–35)
Grade III, n (%)	161 (23%)	50 (20%)	111 (25%)
Positive nodes, median (IQR)	3 (1–7)	3 (1–7)	3 (1–7)
Recurrence or death, n (%)	299 (44%)	94 (38%)	205 (47%)

Women given tamoxifen were older (median 58 vs 50 years) and more often postmenopausal (76% vs 48%); tumor size, grade and nodes were similar (n = 686).

Study at a Glance

Pattern: key numbers

686

patients

Node-positive breast cancer

299

recurrence-or-death events

44% of patients

0.69

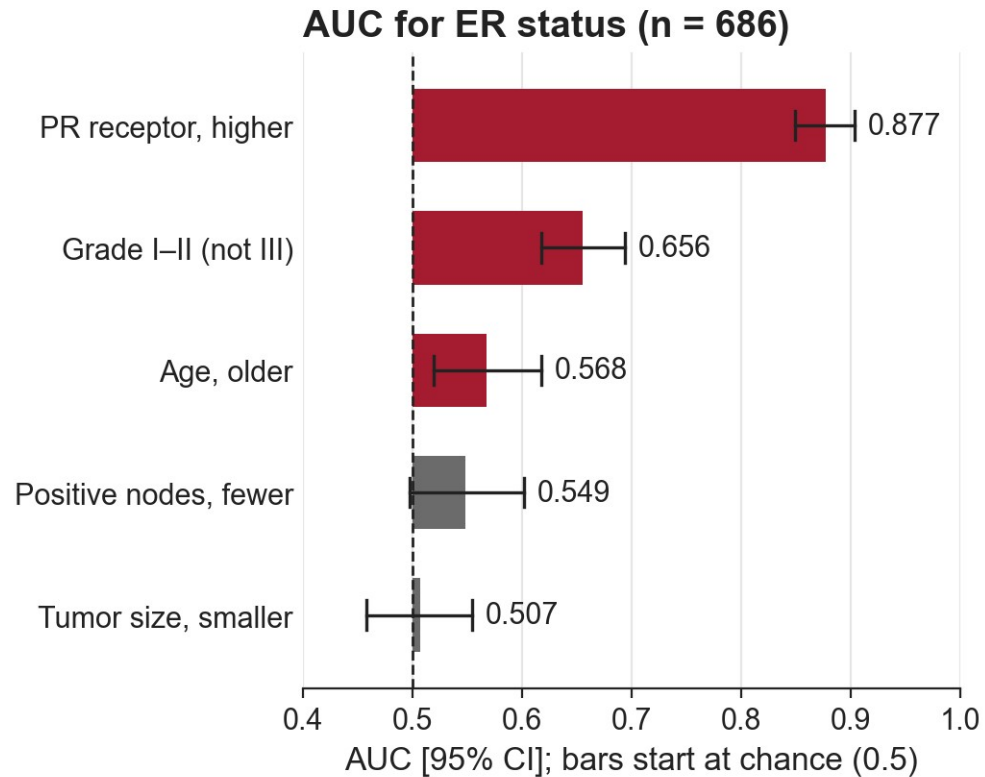
HR, tamoxifen vs none

95% CI 0.54–0.89; $p = 0.004$

Tamoxifen is associated with longer recurrence-free survival (HR 0.69 [0.54–0.89], $p = 0.004$; $n = 686$, 299 events).

Single Predictors of ER Status

Pattern: figure + takeaways

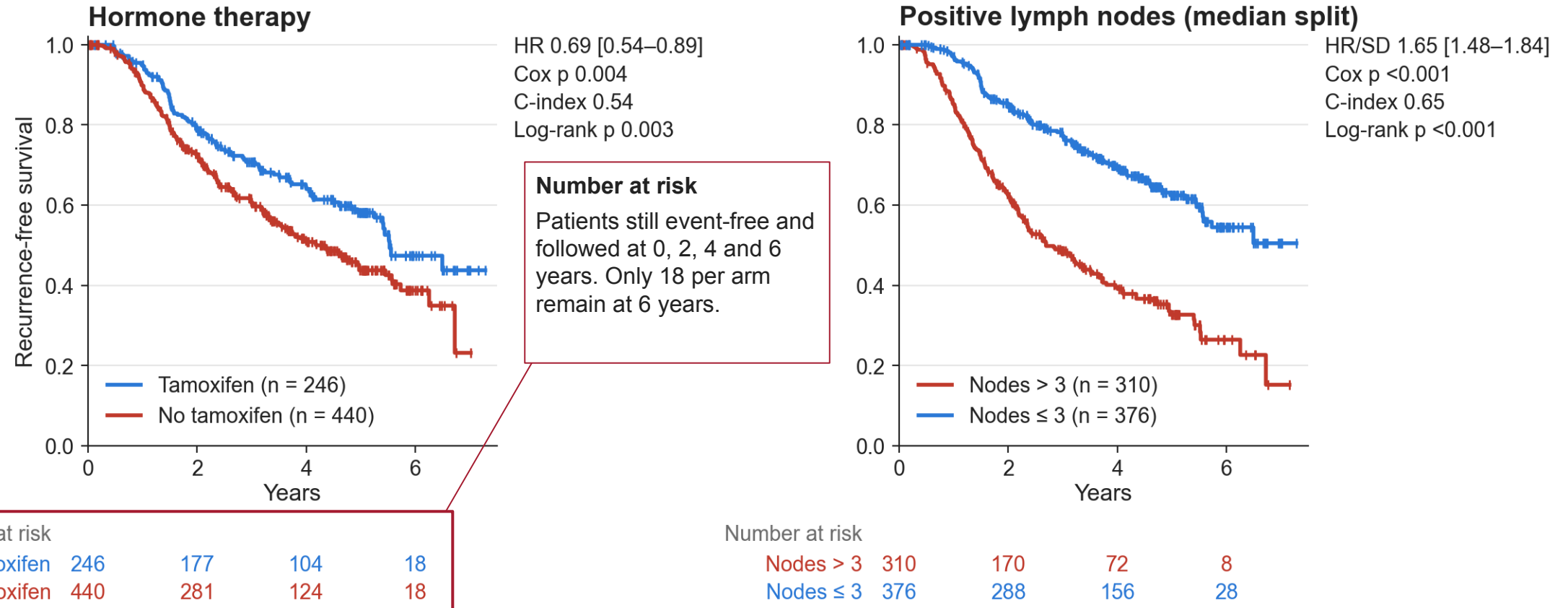


Takeaways

- PR level is by far the strongest (AUC 0.877)
- Grade and age add little (AUC 0.656, 0.568)
- Nodes and tumor size: CIs include 0.5

Of five clinical variables, only PR level discriminates ER status well (AUC 0.877 [0.849–0.904]; 497 of 686 ER-positive).

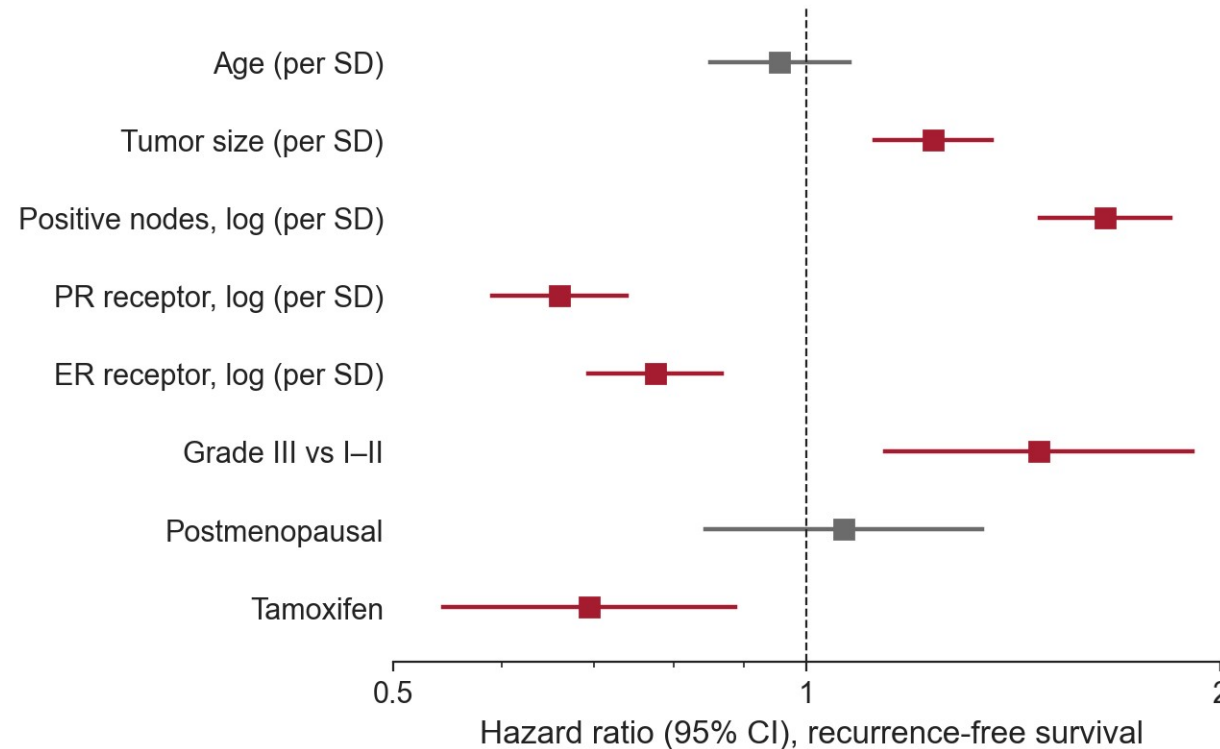
Reading the Kaplan–Meier Panel



By 6 years only 18 women per arm remain at risk, so the curves beyond 6 years rest on few patients (n = 686, 299 events).

Clinical Factors — Univariate Cox

Pattern: progressive emphasis 1/2

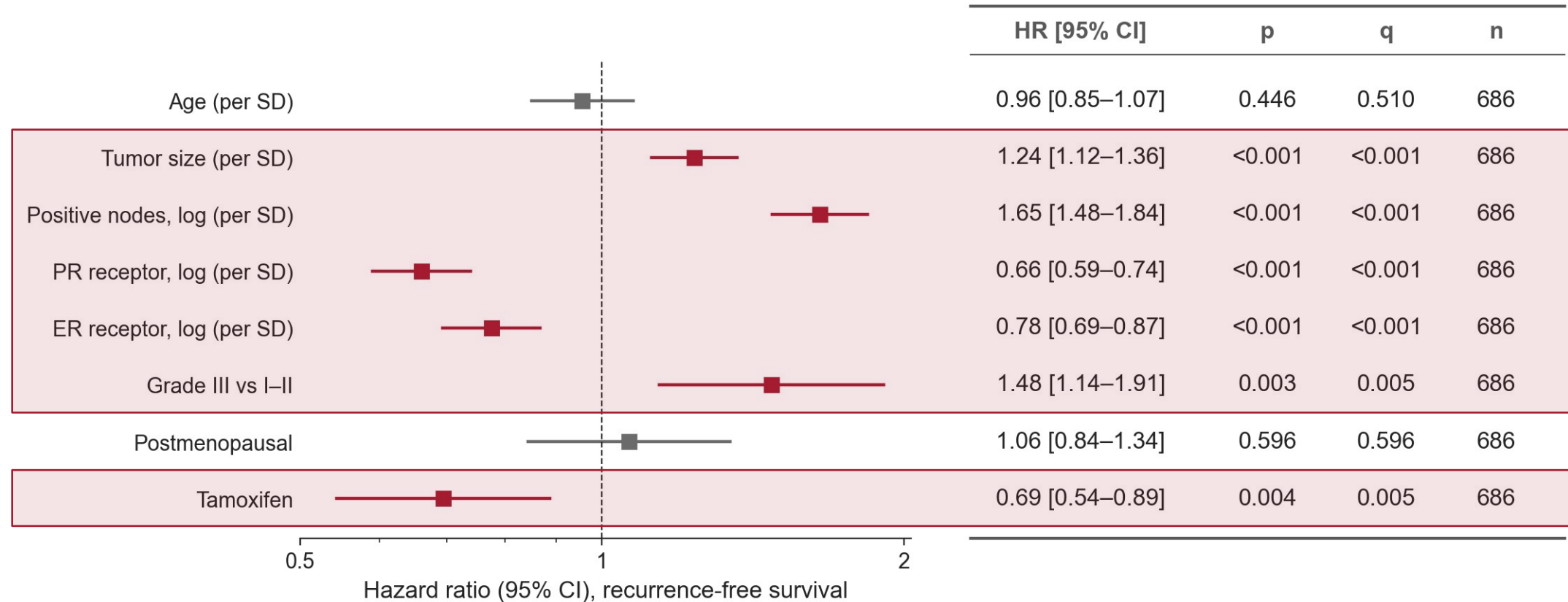


HR [95% CI]	p	q	n
0.96 [0.85–1.07]	0.446	0.510	686
1.24 [1.12–1.36]	<0.001	<0.001	686
1.65 [1.48–1.84]	<0.001	<0.001	686
0.66 [0.59–0.74]	<0.001	<0.001	686
0.78 [0.69–0.87]	<0.001	<0.001	686
1.48 [1.14–1.91]	0.003	0.005	686
1.06 [0.84–1.34]	0.596	0.596	686
0.69 [0.54–0.89]	0.004	0.005	686

Hazard ratios range from 0.66 per SD (PR receptor) to 1.65 per SD (positive nodes) across eight clinical factors (n = 686, 299 events).

Clinical Factors — Univariate Cox

Pattern: progressive emphasis 2/2



Six of eight factors have $q < 0.05$ (all $q \leq 0.005$): larger tumors, more nodes and grade III with shorter, higher receptor levels and tamoxifen with longer recurrence-free survival.

Feature Definitions

Pattern: feature-definition table

Feature	Low → high	What is computed (unit)	Higher value	Lower value	HR > 1 means
Age	21 → 80 years	Age at entry (years) per SD = 10.1 years	Older	Younger	Older recur sooner
Tumor size	3 → 120 mm	Largest diameter (mm) per SD = 14.3 mm	Larger	Smaller	Larger recur sooner
Positive nodes	1 → 51 nodes	$\log(1 + \text{nodes})$ per SD = 0.72	More nodes	Fewer nodes	More nodes recur sooner
PR receptor	0 → 2,380 fmol/mg	$\log(1 + \text{PR, fmol/mg})$ per SD = 1.93	More PR	Less PR	More PR recur sooner
Grade III	I–II → III	Grade III vs I–II yes vs no	Grade III	Grade I–II	Grade III recur sooner

Continuous features are scaled per SD in this cohort (n = 686), so HR > 1 means a one-SD higher value goes with shorter recurrence-free survival.

Model Comparison — C-index

Pattern: model comparison

Cox model	Variables	Apparent C-index	5-fold CV C-index
Positive nodes	1	0.645	0.646
Nodes + size + grade	3	0.655	0.650
All clinical	8	0.698	0.688

All clinical = nodes, tumor size, grade III, age, postmenopausal, PR, ER and tamoxifen. Apparent C-index: fitted and evaluated on the same 686 patients.

Adding all eight clinical variables raises the apparent C-index from 0.645 (nodes only) to 0.698 on the same patients (n = 686, 299 events); 5-fold CV gives 0.688.

Tamoxifen — Adjusted Hazard Ratio

DRAFT – to be confirmed

Pattern: draft

Cox model for tamoxifen	HR [95% CI]	p	n (events)
Univariate	0.69 [0.54–0.89]	0.004	686 (299)
Adjusted for 7 clinical variables	0.68 [0.53–0.88]	0.003	686 (299)

Adjusted for positive nodes, tumor size, grade III, age, postmenopausal status, PR and ER (continuous per SD).

After adjustment for seven clinical variables the tamoxifen HR barely changes (0.68 [0.53–0.88] vs 0.69 unadjusted, p = 0.003; n = 686, 299 events).

HR: hazard ratio; CI: confidence interval; PR: progesterone receptor; ER: estrogen receptor.

Supplementary — Univariate vs Multivariable

Pattern: supplementary (hidden)

Variable	Univariate HR [95% CI]	p	q	Multivariable HR [95% CI]	p
Age (per SD)	0.96 [0.85–1.07]	0.446	0.510	0.92 [0.77–1.10]	0.341
Tumor size (per SD)	1.24 [1.12–1.36]	<0.001	<0.001	1.05 [0.94–1.18]	0.348
Positive nodes, log (per SD)	1.65 [1.48–1.84]	<0.001	<0.001	1.58 [1.41–1.78]	<0.001
PR receptor, log (per SD)	0.66 [0.59–0.74]	<0.001	<0.001	0.67 [0.58–0.78]	<0.001
ER receptor, log (per SD)	0.78 [0.69–0.87]	<0.001	<0.001	1.06 [0.91–1.24]	0.456
Grade III vs I–II	1.48 [1.14–1.91]	0.003	0.005	1.09 [0.83–1.44]	0.524
Postmenopausal	1.06 [0.84–1.34]	0.596	0.596	1.22 [0.86–1.75]	0.266
Tamoxifen	0.69 [0.54–0.89]	0.004	0.005	0.68 [0.53–0.88]	0.003

Nodes, PR receptor and tamoxifen stay associated in the multivariable model (HR 1.58, 0.67, 0.68; all $p \leq 0.003$); tumor size, grade III and ER do not (n = 686, 299 events).